Assignment 7C. BVP – Shooting and Finite element method Marks 20

Posted on 15.10.2026 @ 2:30 pm and due on 15.10.2026 @ 5:40 pm

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1. Consider a cantilever beam of length $L = 2.0$ m, square cross-section $\alpha = 0.05 \times 0.05\,\mathrm{m}^2$, uniformly distributed load $q = 5000\mathrm{N/m}$ and Young's modulus $Y = 2.0 \times 10^{11}\,\mathrm{N/m}^2$. The depression of the beam $y(x)$, when fixed at the two ends, is given by

$$\frac{d^2y}{dx^2} = -\frac{q}{2EI}\,x(x-L)$$

where I is the moment of inertia. Determine the maximum and half-maximum depressions. **[7]**

2. Equation for heat conduction in a thin, un-insulated rod of length $L = 10$ m is

$$\frac{d^2T}{dx^2} + \alpha(T_a - T) = 0$$

where the heat transfer coefficient $\alpha = 0.01\,\mathrm{m}^{-2}$ parameterizes heat dissipated to the surrounding air and $T_a = 20^o\,\mathrm{C}$ is the ambient temperature. If $T(x=0) = 40^o\,\mathrm{C}$ and $T(x=L) = 200^o\,\mathrm{C}$, solve the boundary value problem to determine at what x the temperature is $T = 100^o\,\mathrm{C}$. **[7]**

3. Solve the following steady state heat conduction problem using finite element method

$$y''(x) = -2 \quad \text{where} \quad x \in [0,1] \quad \text{and} \quad y(x=0) = 1,\ y(x=1) = 3$$

Compare your solution with the analytical result $y(x) = -x^2 + 3x + 1$. **[6]**